



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year: 2024- 2025 (Odd Semester)

Degree, Semester & Branch: IV Semester B.E. CSE

Course Code & Title: CB3491- Cryptography and Cyber Security

Name of the Faculty member (s): Mrs.K. Jeyakarthika

Innovative Practice Description

- **Unit / Topic:** Unit V / Network Access Control
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO 18
- **Activity Chosen:** Learning by Teaching
- **Justification:**

The main objective of the learning by teaching activity was to promote active learning and knowledge consolidation among the students. By assigning topics to the students and having them teach those topics to their peers, the activity aimed to reinforce their understanding, improve communication skills, and foster a collaborative learning environment.

- **Time Allotted for the Activity:** 40 minutes
- **Details of the Implementation:**
 - Before the class, the teacher, inquired about the students' interest in teaching the concept of Network Access Control.
 - Arun and Aswath P of III-year CSE expressed interest in taking the class shown in Figure 1.
 - The topic “Network Access Control” was assigned to the student.
 - The adequately time was given to the student to prepare for the assigned topic and encouraged to refer more materials from various sources.
 - To keep the class interested, student utilized a variety of instructional strategies.
 - To promote involvement, students were allowed to ask questions to their peers during the presentation. Because of this, students were able to have conversations and share their perspectives about topic which very important in programming.
 - Finally, the instructor summarized the content.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO12	PSO1	PSO2	PSO3
CO5	3	2	3	2	3	1	2	2	3	2	2

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO5
	3	3	3	3	3
Justification for correlation	Apply the basic Computer Science Engineering and mathematics Knowledge for implementing cybercrime and Information technology.	Analyze various digital signature schemes are highly correlated.	Developing systems with basic digital signature mechanisms are highly correlated	Investigate various aspects of cryptanalysis is highly correlated.	Use of virtual environment, tools for performing password cracking, SQL injection is highly correlated.
	PO8	PO9	PO12		
	1	2	1		
	To complete the assignments/ tasks in an ethical way.	The students will able to work as an individual and team to how to distribute a key with integrity and authentication.	Understanding concepts of cybercrimes in cyber security for competing future demands		

- **Images / Screenshot of the practice:**



Figure 1: Teaching the concept of Network Access control

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- All the students were eagerly listened the class. Students felt they are able to understand the topic.
- The students were able to remember key details and concepts more easily.

- Students found the seminar activity to be a valuable learning experience that enhanced their understanding, communication skills and confidence.
- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
 - By involving students in teaching and discussions, the seminar promotes active learning, making the educational experience more dynamic and interactive.
 - This activity helped the student to improve communication skills.
 - Students spent their time in self-learning.
 - This activity encouraged the students to share their knowledge with others.
- ❖ **Challenges faced in implementation:**
 - Students are not participating in the activity due to shyness, fear of public speaking, or lack of confidence.
 - Implementing Learning by Teaching required additional time compared to traditional teaching methods.
 - Planning and executing interactive teaching sessions was challenging.

References:

1. <https://effectiviology.com/protege-effect-learn-by-teaching>
2. https://www.researchgate.net/publication/351905566_Impact_of_Seminars_on_Student_Soft_Skills_Development
3. https://www.ritrjpm.ac.in/images/computer-science/2023-2024/IP/LBT_CD_IGA23-24.pdf